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Autosomal Allelic Inactivation: Variable Replication and Dosage Sensitivity

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eLife Assessment

This **important** study links allelic expression imbalance with replication timing, suggesting a stochastic model for haploinsufficiency in dosage-sensitive disease. The integration of allele-specific RNA-seq and replication timing in clonal systems provides **solid** evidence for an association between asynchronous replication and allelic imbalance, although the scope and generality of some conclusions require more cautious interpretation. This study will interest epigeneticists and genome regulation researchers studying replication timing and monoallelic expression, as well as developmental biologists and human geneticists concerned with clonal heterogeneity, haploinsufficiency, and variable disease penetrance.

[Editors' note: this paper was reviewed by *Review Commons* [✉](#).]

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Abstract

Autosomal monoallelic gene expression and asynchronous replication between alleles are well-established features of imprinted genes and genes regulated by allelic exclusion. Inactivation/Stability Centers (I/SCs) are recently described autosomal loci that exhibit epigenetic regulation of allelic expression and replication timing, with differences that can be comparable to those observed between the active and inactive X chromosomes¹. Here we characterize hundreds of autosomal loci with allele-specific epigenetic regulation of replication timing and gene expression, defining them as I/SCs. I/SCs are approximately 1 megabase in size and can contain both protein-coding and noncoding genes. In different single cell derived clones, these genes may be expressed from a single allele, the opposite allele, both alleles, or not expressed at all. This stochastic, yet mitotically stable, pattern indicates that the choice of which allele is expressed is independent of parent of origin and independent of the expression status of the other allele. Similarly, alleles within I/SCs show varying replication timing, either earlier or later, that is also independent of the other allele. Additionally, we find corresponding I/SCs in the mouse genome that display conserved synteny with human I/SCs. This allele-restricted regulation creates extensive cellular mosaicism through a stable epigenetic mechanism. This mosaicism impacts numerous dosage-sensitive genes associated with human diseases such as Parkinson, epilepsy, deafness, and impaired intellectual development.

Introduction

The relationship between monoallelic expression and asynchronous replication between alleles of autosomal genes, exemplified by genes subject to genomic imprinting or allelic exclusion, has been the focus of extensive research for over three decades^{2–4}. In recent work, we employed haplotype-phased RNA expression and replication timing assays on multiple single-cell-derived Lymphoblastoid Cell Line (LCL) clones from two unrelated individuals to identify loci exhibiting allele-restricted expression and replication timing¹. This clonal, allele-specific analysis across autosomes revealed an unexpected genomic feature at approximately 200 loci, exhibiting allele-restricted gene expression (including random monoallelic expression); and variable replication timing (including asynchronous replication between alleles). The magnitude of allelic differences in expression and replication timing observed in some of the clones was comparable to the large differences detected between the active and inactive X chromosomes. We have named these autosomal loci Inactivation/Stability Centers (I/SCs)^{1,5–7}.

I/SCs can encompass both coding and noncoding genes that may be expressed from a single allele in some clones, expressed from the opposite allele in other clones, biallelically expressed in other clones, or silent on both alleles in yet other clones. This pattern of allele-restricted expression indicates that each allele independently adopts either an expressed or silent state. Importantly, because these expression states are mitotically stable, allele-autonomous, and independent of parental origin, we refer to the choice of the expressed allele as stochastic¹. To describe this phenomenon, we and others have used the term Allelic Expression Imbalance (AEI) to refer to the stochastic nature of autosomal random monoallelic expression^{1,8–11}. Additionally, the replication timing profiles on autosomes are highly variable and clone-specific, exhibiting both synchronous and asynchronous patterns in different clones. We refer to this as Variable Epigenetic Replication Timing (VERT)¹.

Random monoallelic expression in the olfactory, immune, and central nervous systems is well established and plays a critical role in generating cell-specific identities^{12–14}. However, increasing evidence indicates that autosomal genes outside these systems also exhibit allele-specific expression, suggesting that AEI plays a broader role across diverse cell types^{4,13}. The functional diversity of these genes implies that AEI contributes to a broad array of cellular processes and is not limited to specialized systems.

In the present study, we aimed to determine whether I/SCs could be detected in the genomes of clonal populations of human primary cells derived from a tissue distinct from those previously examined. To this end, we employed articular cartilage progenitor (ACP) cells as a second model system¹⁵. ACP cells are non-transformed yet are particularly well-suited for clonal analysis due to several advantages: 1) they can be readily isolated from amputated tissue of patients with polydactyly, 2) they exhibit high proliferative capacity, and 3) they can be efficiently expanded from single cells¹⁶. We identify an additional 163 VERT regions in the human genome, increasing the total number of potential I/SCs to 363, which collectively cover approximately 12% of the human genome. In addition, we found that many previously reported “random monoallelic” genes, including those encoding olfactory and vomeronasal receptors^{3,17}, antigen receptors¹⁸, and homophilic cell adhesion proteins¹², are located within VERT regions. Our findings confirm that these well-established “random monoallelic” genes reside within genomic loci that exhibit the epigenetic hallmarks, namely allele-restricted expression and variable replication timing, that define I/SCs.

The relationship between genotype and phenotype is central to understanding human genetic disease. While most loss-of-function mutations are recessive, indicating that a single normal allele is sufficient for typical function, dominantly inherited disorders often result from gain-of-function mutations. However, a significant fraction of dominant diseases arises through haploinsufficiency (i.e. heterozygous for a loss of function mutation), where a single functional allele fails to maintain normal physiology. Although haploinsufficiency may affect up to 40% of human genes^{19–21}, the underlying molecular mechanisms remain poorly understood. Current models suggest that insufficient gene dosage may arise from a lack of redundancy, critical expression thresholds,

stoichiometric imbalances, or toxicity associated with a lack of overexpression buffering ^{22–25}. In contrast, insights from X-linked disease inheritance illustrate how cellular mosaicism and cellular selection can shape phenotypic outcomes: in XX individuals, differences in the survival or expansion of cells expressing either the wild-type or mutant alleles can influence whether a disease manifests as recessive or dominant ^{21,26,27}.

The magnitude of AEI is often quantified using allelic expression ratios, which represent the proportion of transcripts derived from each allele. Importantly, recent findings have demonstrated that even modest allelic expression biases can have functional consequences, particularly in disease contexts. One recent study directly linked small but stable AEI to variable clinical outcomes among individuals carrying the same pathogenic mutation, underscoring AEI as a potential modulator of disease expressivity ²⁸. These observations highlight the importance of considering AEI as a disease modifier that may shape phenotypic diversity and influence the penetrance of genetic disorders.

Here, we demonstrate that numerous genes implicated in autosomal human genetic diseases, such as Parkinson (DNAJC6, LRRK2 and SNCA), epilepsy (SCN1A, GABR1, and SAMD12), deafness (ATP11A, EPS8, and MYO6), and neurodevelopmental disorders (KCNA1, RETREG1, ROBO1, and TIAM1), are located within VERT regions and display AEI. Furthermore, mapping I/SCs across all autosomes reveals 87 known dominantly inherited single-gene disorders that map within VERT regions, where loss of function mutations in most of these genes result in haploinsufficiency ²⁹. We propose that epigenetically regulated stochastic AEI can contribute to pathogenic phenotypes in heterozygous individuals by either reducing the number of functional cells below a critical disease threshold, akin to X linked dominant disorders, or by disrupting the normal mosaicism of gene expression within disease-relevant tissues.

Results

Here we employed articular cartilage progenitor (ACP) cells¹⁵, as a distinct cell type from those from our previous studies, as a second model system to identify I/SCs. For this analysis, we isolated ACP cells from amputated digits of patients with polydactyly. In addition, we sequenced genomic DNA from the parents of the patients with polydactyly to generate phased genomes of the ACP cells. We analyzed two sets of ACP clones isolated from two unrelated individuals, one female (ACP7) and one male (ACP6), for haplotype-phased Repliseq and RNAseq (Fig. 1b [↗](#)). Analysis of the female ACP7 clones allowed us to establish the statistical parameters for the assignment of VERT and AEI present in the ACP clones by directly comparing the magnitude of autosomal allele-specific replication timing and RNA expression to that of the active and inactive X chromosomes present in each clone (Fig. 1c [↗](#); and see Supplementary Tables 1, 2 and 3). Taken together the combined VERT and AEI analysis allowed us to characterize additional I/SCs in the human genome. An illustration of a hypothetical I/SC, with the region of VERT shaded in gray and the expression of coding and noncoding genes indicated as stippled or smooth rectangles, respectively, is shown in Figure 1d [↗](#).

To identify genomic regions with epigenetic variation between alleles, we calculated the standard deviation (SD) of replication timing across alleles for each 250 kb genomic window. Windows with an SD value ≥ 2.5 standard deviations above the genome-wide median were classified as outliers. This analysis identified 163 additional VERT regions (Supplementary Table 1). Figure 1e [↗](#) illustrates this analysis for both ACP (ACP6 and ACP7) and LCL (EB3-2 and GM12878) clones on Chromosome 1, where windows are indicated as gray circles and outliers indicted in different colors.

To detect coding genes with AEI in ACP clones, we defined AEI as differential expression between alleles using recently established criteria ^{1,28}. Genes were considered AEI-positive if they exhibited $\geq 80\%$ allelic imbalance ($\text{AEI} \geq 0.80$ or ≤ 0.20) with $\text{FDR} \leq 0.05$ in at least one clone. Additional analyses using more stringent thresholds ($\text{AEI} \geq 0.90$ or ≥ 0.95) are provided in Supplementary Table 2. To exclude imprinted genes and effects due to genetic polymorphisms (i.e. expression Quantitative Trait Loci, eQTL³⁰) we required that each AEI gene also be biallelically expressed (AEI

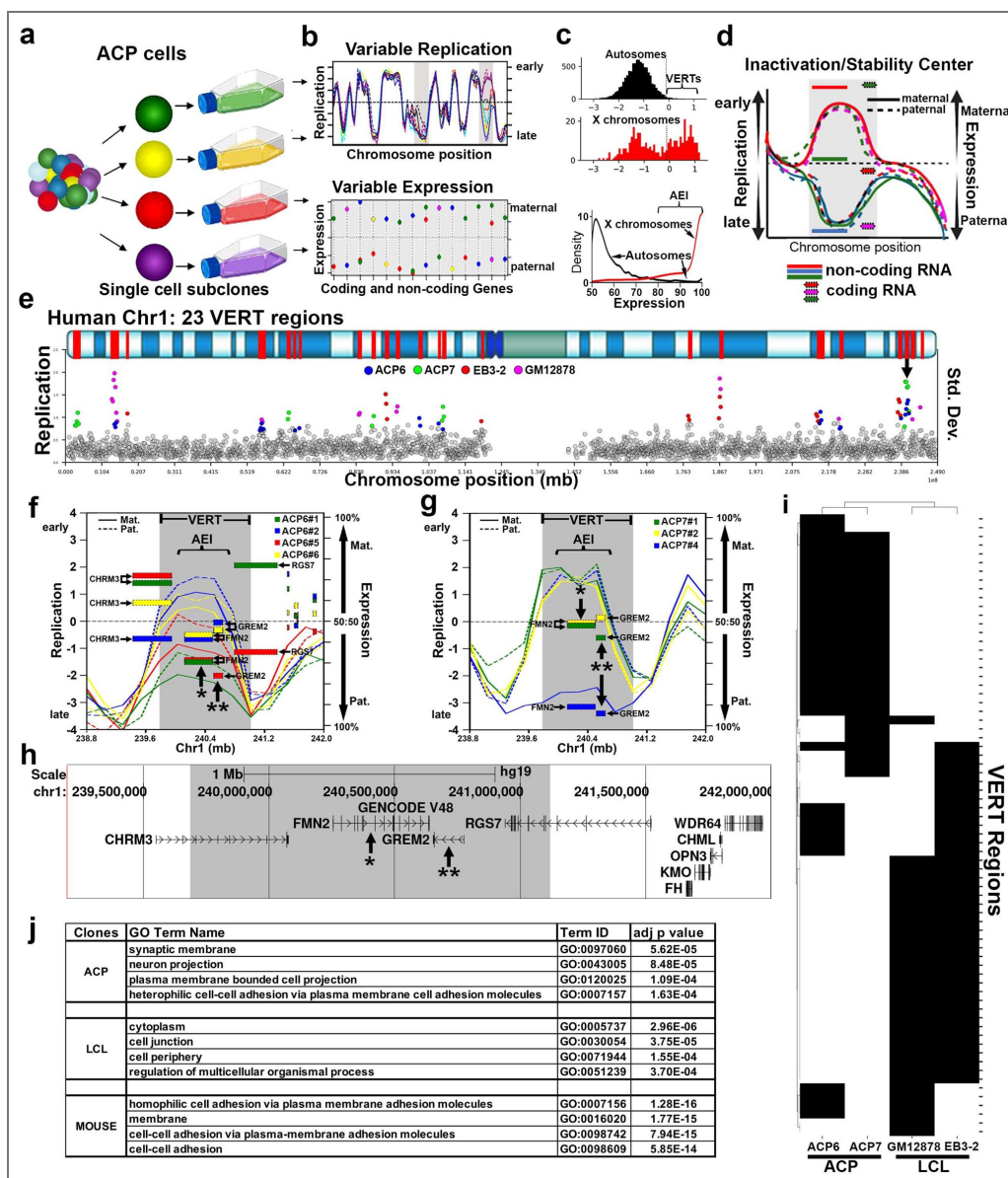


Figure 1. Genome wide variable replication-timing and expression.

a) Single cell derived ACP clones (color coded) are isolated. The maternal and paternal haplotypes were used to analyze allele-specific replication timing and expression. b) The top panel illustrates the expected replication timing profile in multiple clones across an autosomal region. The regions that display VERT are highlighted by shading. The bottom panel illustrates the expected expression variability in different clones from the same individual. c) We define AEI as an allelic bias that is $\geq 80\%$ allelic imbalance ($AEI \geq 0.80$ or ≤ 0.20). d) Illustration of an Inactivation/Stability Center showing the clonal and allele-restricted variability in replication timing and expression of both protein coding and noncoding genes. e) The standard deviation (Std. Dev.) in 250 kb windows (circles) across human chromosome 1. Outlier windows from all four sets of clones (ACP6, ACP7, EB3-2, and GM12878) are highlighted in different colors as shown. The top panel illustrates the G-banding (blue shading) pattern for human chromosome 1, with 23 VERT regions highlighted in red. The arrow marks the VERT region shown in panels f-h below. f-h) I/SC located between 239 and 242 mb of chromosome 1 (see panel e). f) VERT (shaded region) and AEI of genes (rectangles) detected in the ACP6 clones, each clone was color coded as shown. For replication, the solid lines represent the maternal (Mat.) allele and the dotted line represents the paternal (Pat.) allele. g) VERT (shaded region) and AEI of genes (rectangles) detected in the ACP7 clones, each clone was color coded as shown. h) UCSC Genome Browser view of the genomic region in panels f and g. The shaded area highlights the VERT region. i) Hierarchical clustering of the VERT regions that are shared between 2 or more clone sets showing both ACP and LCL clones. j) Gene Ontology (GO) enrichment analysis of the genes located within the VERT regions, the most significant hits for each set of clones, organized by cell type, are shown. GO analysis of the mouse genes located within the pre-B cell clone VERT regions are also shown.

< 0.70) in at least one clone from the same individual with ≥ 20 informative reads. Using these criteria, we identified 256 (2.4% of expressed genes) autosomal coding genes exhibiting AEI across the two ACP clone sets (Supplementary Table 2).

In addition, we also sought to identify AEI of extremely long noncoding RNAs (i.e. ASAR lincRNAs [1,5,7,31](#)) expressed in the ACP clones. For this analysis we refer to regions of the genome with >50 kb of contiguous transcription of noncoding DNA as Transcribed Loci (TL) [1](#). Using the same criteria used for the coding genes described above, we identified 23 TLs exhibiting AEI across the two ACP clone sets (Supplementary Table 3).

Allelic Expression Imbalance and Variable Epigenetic Replication Timing define Inactivation/Stability Centers

By integrating the location of genes exhibiting AEI with genomic regions displaying VERT, we identified overlapping loci that define I/SCs. [Figures 1f](#) and [1g](#) illustrate a representative I/SC located on chromosome 1 at approximately 240 Mb (marked by an arrow in [Fig. 1e](#)). Notably, VERT was detected at this locus across both sets of ACP clones ([Fig. 1f](#) and [1g](#)). [Figure 1h](#) presents the corresponding UCSC Genome Browser view of this locus, with genes showing AEI marked by asterisks (see Supplemental Table 2). Additional examples of I/SCs located on chromosomes 18, 6 and 9 are shown in [Figure 2](#).

One of the primary objectives of this study was to determine whether different tissues utilize distinct sets of I/SCs, as might be expected given that different cell types express unique gene repertoires that could be subject to AEI. To explore this, we performed hierarchical clustering analysis of the VERT regions identified in the two sets of ACP clones and the two sets of LCL clones. [Figure 1i](#) shows this analysis and reveals tissue-specific patterns with a relatively small degree of overlap in VERT regions across the two tissue types: ACP6 clones share more VERT regions with ACP7 clones than with either LCL clone sets, and the reverse is true for the LCL clones. These findings indicate that while some I/SCs are shared across tissues, there is a large component of tissue-specific epigenetic regulation of allelic replication timing.

At the molecular level, stochastic monoallelic expression contributes to cellular individuality and introduces functional variability among individual cells within complex biological systems [12–14](#). This mechanism is particularly well characterized in the olfactory, immune, and central nervous systems, where the expression of diverse receptor-type molecules through random monoallelic expression underlies cell-specific functional identities. To further investigate the functional characteristics of protein-coding genes located within I/SCs, we annotated the genes residing within the VERT regions identified across all four clonal datasets (ACP clones: ACP6 and ACP7; and LCL clones: EB3-2, and GM12878; see Supplementary Table 4). To assess whether these genes are functionally related, we performed a Gene Ontology (GO) enrichment analysis. This analysis revealed a significant enrichment for genes involved in plasma membrane localization, cell-cell adhesion and neurodevelopmental processes, the most significant hits for each set of clones is presented in [Figure 1k](#) (see full details in Supplementary Table 5). This analysis suggests that I/SC-associated epigenetic regulation may play a broader role in the establishment of cellular identity and tissue organization particularly in the nervous system.

Gene clusters with AEI map to VERT regions

In the olfactory, immune, and central nervous systems, random monoallelic expression of gene family members, often organized in genomic clusters, plays a key role in establishing cell-specific identities [12–14](#). Consistent with these observations, we found that VERT regions frequently coincide with gene clusters in the human genome (Table 1). For example, [Figures 3a](#) and [3b](#) highlight a cluster of 11 olfactory receptor (OR) genes located on human chromosome 19 at approximately 15 Mb. This region also harbors a cluster of Cytochrome P450 (CYP4F family, $n = 6$) genes. The interspersed genomic arrangement of the CYP4F genes among the OR genes, indicates that both gene families reside within the same I/SC and therefore both may be subject to AEI.



a, d and g) VERT regions on human chromosomes 18, 6 and 9, respectively. The top panels illustrate the G-banding (blue shading) pattern for each human chromosome, with VERT regions highlighted in red. The standard deviation (Std. Dev.) in 250 kb windows (circles) across each human chromosome is shown. Outlier windows from all four sets of clones (ACP6, ACP7, EB3-2, and GM12878) are highlighted in different colors as shown. The arrows and red boxes mark the VERT regions shown in panels b, e, and h. b, e and h) 1/SCs highlighted in panels a, d, and g. For replication, the solid lines represent the maternal (Mat.) allele and the dotted line represents the paternal (Pat.) allele. The regions with VERT are shaded and the AEI is shown for each clone in a different color. c, f and i) UCSC Genome Browser views of the genomic regions in panels b, e and h. The shaded areas highlight the VERT regions. Genes with AEI are marked with +, *, or #. b and c) The + marks MOCOS, which shows AEI in ACP cells (Supplemental Table 2) and ++ marks FHOD3, which shows AEI in LCLs¹. e and f) The * marks from DNAH8, ** marks transcripts from KCNK5, and *** marks non-coding transcripts from ENSG00000293066 (also named: TL:6-39.9.1¹). All three of these genes show AEI in LCL clones¹. h and i) The # marks transcripts from DAPK1, ## marks transcripts from SPIN1. Both genes show AEI in LCL clones¹.

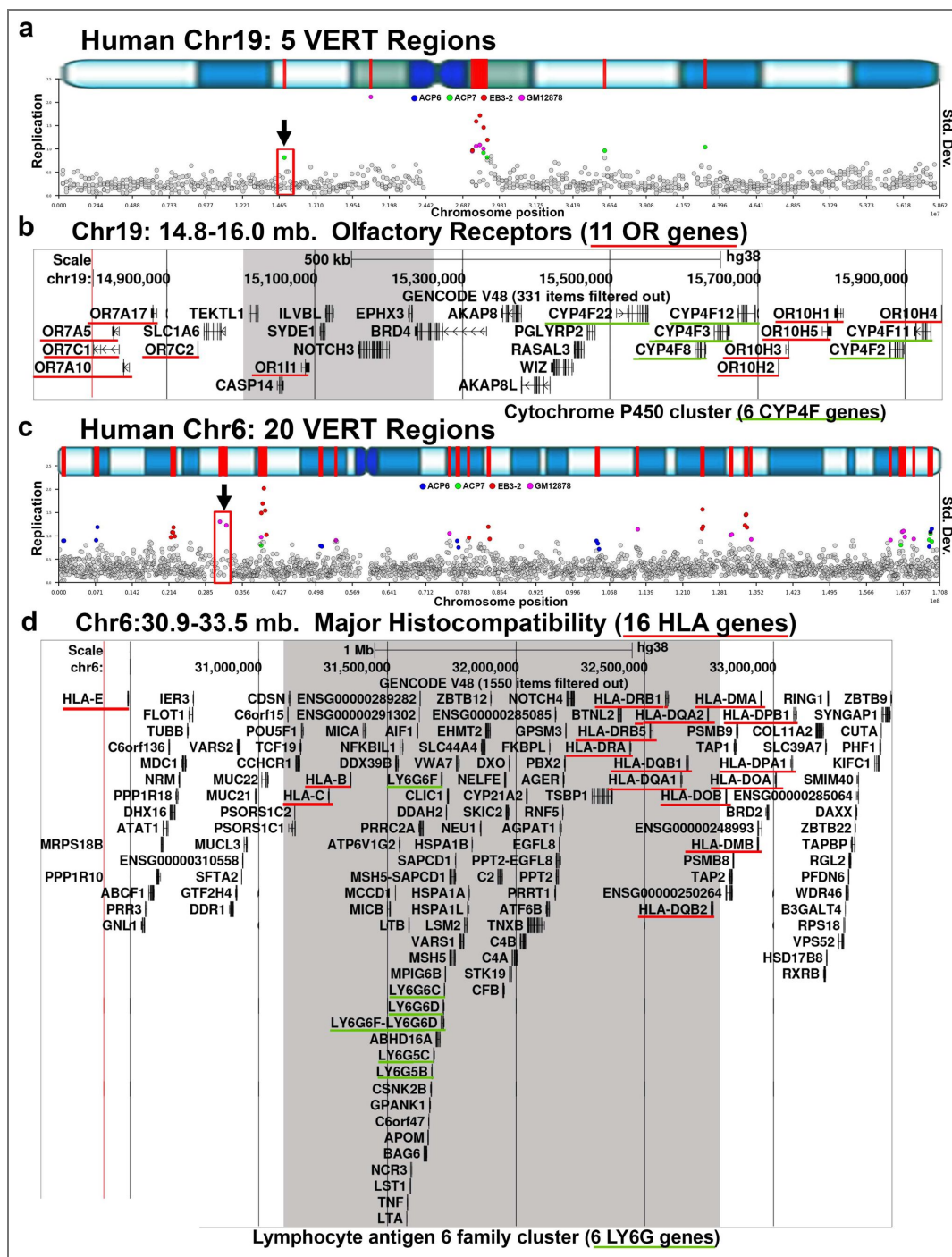


Figure 3. Human gene clusters with AEI map to VERT regions.

a and c) VERT regions on human chromosome 9 and 6. The top panels illustrate the G-banding (blue shading) pattern for each human chromosome, with VERT regions highlighted in red. The standard deviation (Std. Dev.) in 250 kb windows (circles) across each human chromosome is shown. Outlier windows from all four sets of clones (ACP6, ACP7, EB3-2, and GM12878) are highlighted in different colors as shown. The arrows and red boxes mark the VERT regions shown in panels b, e, and h. The red box and arrow identify the I/SC in panel b. b) UCSC genome browser view illustrating the genomic location highlighted in panel a. The shaded area highlights the VERT region. The red lines mark 11 olfactory receptor genes, and the green lines mark 6 cytochrome P450 family (CYP4F) genes. d) UCSC Genome Browser view illustrating the genomic location highlighted in panel c. The shaded area highlights the VERT region. The red lines mark 16 HLA genes, and the green lines mark 6 Lymphocyte antigen 6 family (LY6G) genes.

A second example of a gene cluster is shown in [Figures 3c](#) and [3d](#), where the major histocompatibility complex (MHC), comprising 16 HLA genes (which are known to be subject to AEI [28,32](#)), lies within a VERT region. Notably, this locus also contains a cluster of Lymphocyte Antigen 6 genes (LY6G family, $n = 6$), which are interspersed among the HLA genes. This genome organization indicates that the LY6G genes, like their neighboring HLA genes, are within the same I/SC and suggests that they too may be subject to stochastic allelic expression.

Mapping mouse VERTs identifies known AEI gene clusters and syntenic I/SCs

Much of our current understanding of stochastic autosomal allelic expression is derived from studies conducted in the mouse [4,9–11,17,18,33–40](#). To assess whether VERT regions could be identified at expected loci in the mouse genome, specifically at sites of known AEI, we analyzed previously published Repli-seq data from mouse pre-B cell clones reported by Blumenfeld et al [41](#). This analysis revealed 117 loci in the mouse genome that display VERT (Supplemental Table 6).

Consistent with our findings in the human genome, the VERT regions detected in the mouse genome frequently overlap with gene clusters, including 5 olfactory receptor gene clusters located on four different autosomes (Table 1). [Figure 4](#) presents three examples: two olfactory receptor (Or) gene clusters on mouse chromosome 2 ([Figs. 4a–4c](#)) and the protocadherin (Pcdh) gene cluster, which is known to generate cellular individuality in the central nervous system through stochastic expression of different Pcdh family members [12,42](#), located on mouse chromosome 18 ([Figs. 4d–4e](#)). In addition to the gene clusters with VERT, we also detected immunoglobulin and T cell receptor gene loci with VERT. As detailed in Supplemental Figure 1, we detected VERT at the T cell receptor alpha (Tra) locus on mouse chromosome 14 at approximately 54 Mb. One striking feature of this locus is that it also contains an olfactory receptor cluster, containing 37 Or genes, and a vomeronasal receptor gene (Vmn2r88). All of these multigene families are documented sites of AEI [3,12,17,42,43](#), highlighting the genomic organization and conservation of I/SC-associated regulation at known AEI genes in the mouse. Gene Ontology analysis of the mouse genes located within these regions showed significant enrichment for biological processes related to the plasma membrane and cell-cell adhesion ([Fig. 1k](#)), which is consistent with the human VERT gene dataset and reinforces the conservation of this regulatory mechanism across species.

Previous studies analyzed a relatively small number of mouse AEI genes to model their human counterparts with respect to autosomal-dominant disorders [38,44](#). To further investigate the conservation of I/SC regulation between the human and mouse genomes, we identified VERT regions that map to syntenic loci across the two species. This analysis revealed 21 loci where VERT is detected in both genomes, as detailed in Table 2. [Figure 5](#) presents three representative examples of syntenic VERT regions: two loci on the short arm of human chromosome 5 with corresponding regions on mouse chromosome 15 ([Figs. 5a–5g](#)), and a third locus on human chromosome 16 with its syntenic region on mouse chromosome 16 ([Figs. 5h–5k](#)). Additional examples of syntenic loci showing AEI and/or VERT, including mouse App (see Gimelbrant et al. [45](#) for AEI of human APP), Mocos and Fhod3 (See [Fig. 2A](#) for VERT and AEI of human MOCOS and FHOD3), and a cluster of 5 sodium channel genes, Scn1a, Scn2a, Scn3a, Scn7a and Scn9a (see [Fig. 6a–c](#) below for VERT and AEI of the syntenic human SCN gene cluster) are shown in Supplemental Figure 2.

We point out that many of the mouse genes that are located within VERT regions are associated with various human genetic diseases (Supplemental Table 7). One of the loci highlighted in [Figure 5a–5d](#) labeled “hi”, exhibits both VERT and AEI ([Fig. 5c](#) and [1](#)), affecting the protein-coding gene SEMA5A, a gene implicated in autism and neurodevelopmental disorders [46–48](#). Notably, this locus also contains a syntenic pair of taste receptor genes: TAS2R1 in humans and Tas2r119 in mice. These genes are part of the taste receptor family, which are known to harbor chromatin marks characteristic of AEI-regulated loci [49](#). Consistent with this observation, we also detected VERT at a taste receptor gene cluster (Tas2r) on mouse chromosome 6 (Table 1).

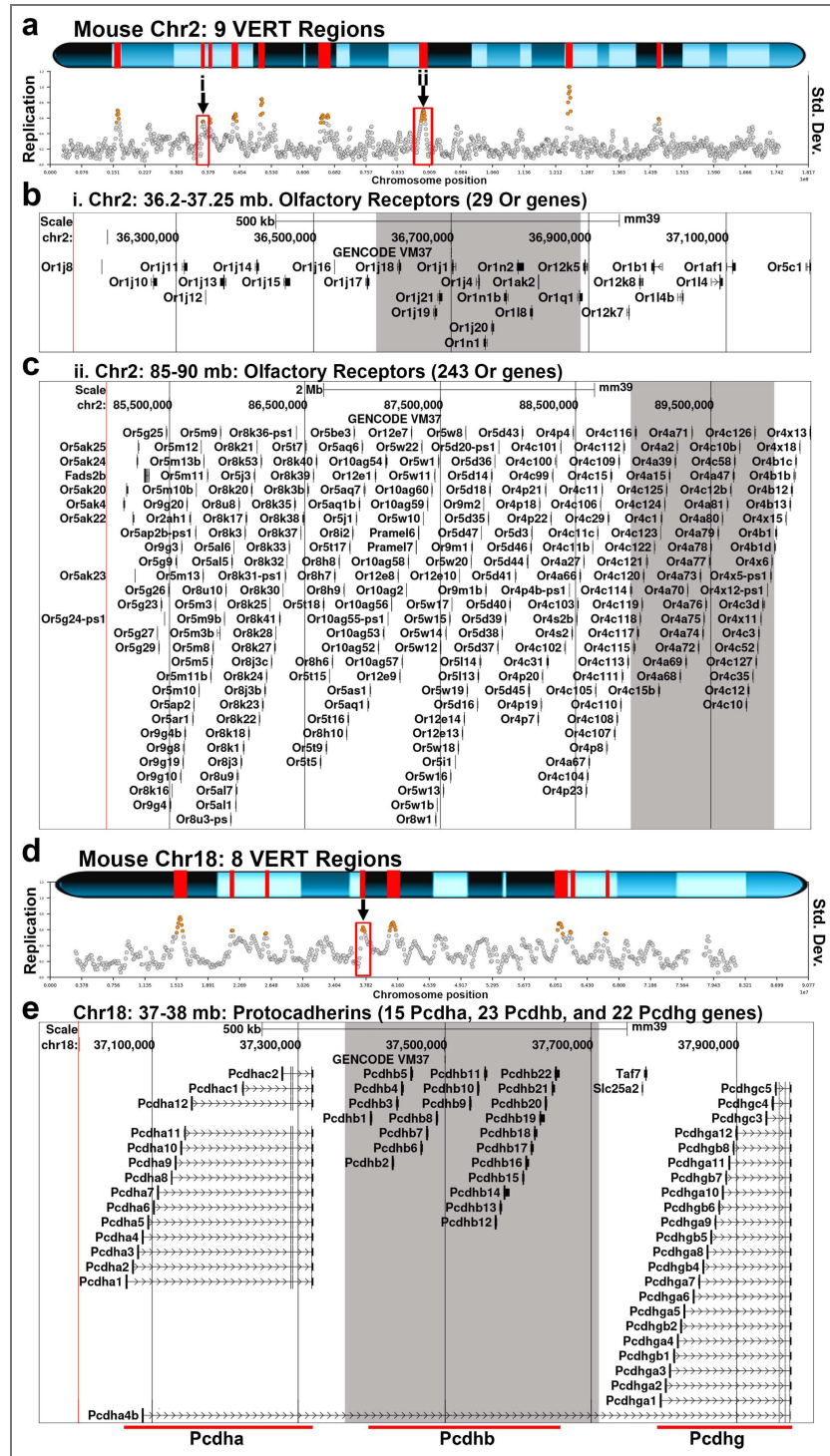


Figure 4. Mouse gene clusters with AEI map to VERT regions.

a and d) VERT regions on mouse chromosome 2 and 18. The standard deviation (Std. Dev.) in 50 kb windows (circles) across each mouse chromosome are shown. Outlier windows from pre-B cell clones are highlighted in orange. The red boxes and arrows identify the VERT regions in panels b (i) and c (ii). b and c) UCSC genome browser views illustrating the genomic locations represented by i and ii in panel a. b) Genome Browser view of the VERT region showing the location of 29 olfactory receptor genes (Or genes). The shaded area represents the VERT region. c) Genome Browser view of the VERT region in ii above, showing the location of 242 olfactory receptor genes (Or genes). The shaded area represents the VERT region. e) Genome Browser view of the VERT region in d above, showing the location of clustered Protocadherin genes (15 Pcdha, 23 Pcdhb, and 22 Pcdhg genes). The shaded area represents the VERT region.

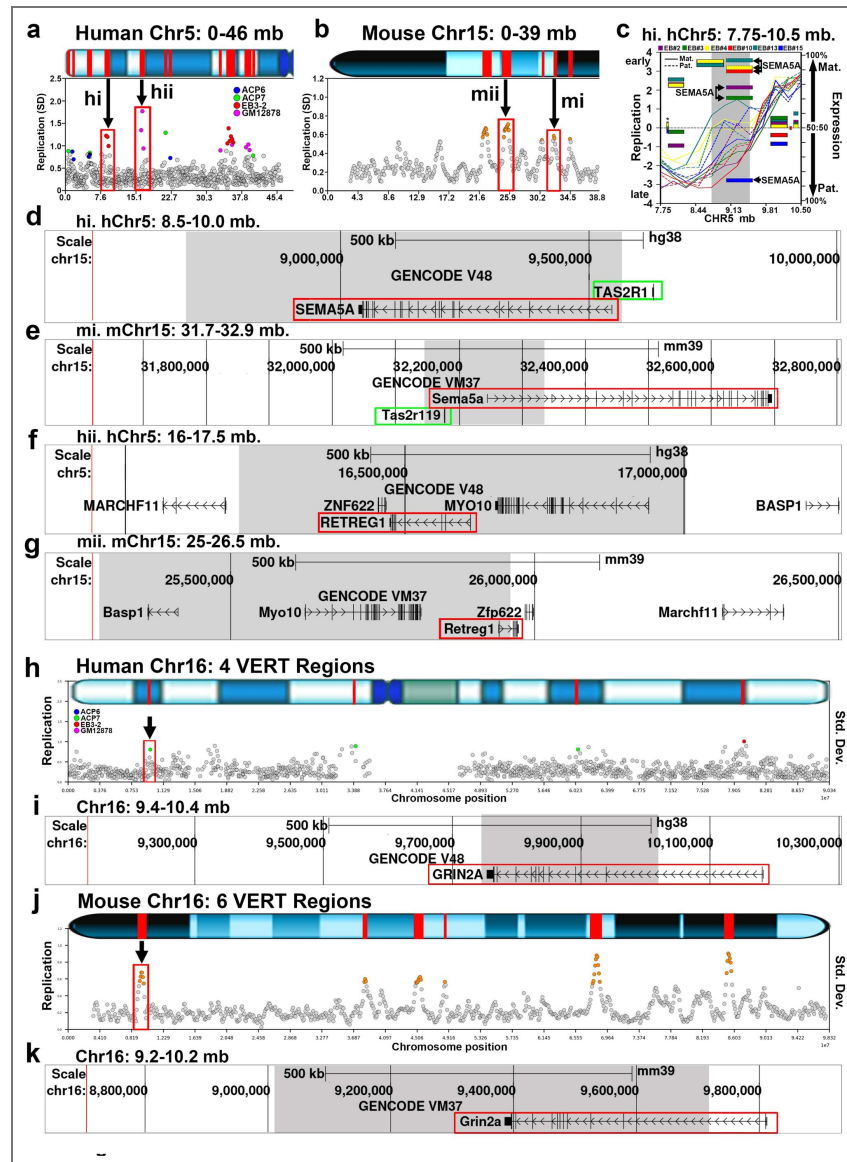


Figure 5. Human and Mouse syntenic regions display VERT.

a and b) The short arm of human chromosome 5, with 11 VERT regions (panel a) and the centromeric region of mouse chromosome 15, with 4 VERT regions (panel b) are shown. a) Two human VERT regions are highlighted, hi and hii, and are syntenic with locations on mouse chromosome 15, mi and mii, that also show VERT (panel b). Note that the gene order in both syntenic regions are inverted with respect to the telomeres when comparing the human and mouse loci. c) Illustration of the I/S on human chromosome 5 between 7.75 and 10.5 mb (hi). The shaded area marks the VERT region, and AEI of the coding gene SEMA5A in different clones is highlighted. For replication, the maternal (Mat.) and paternal (Pat.) alleles are indicated. d) UCSC Genome Browser view of the human VERT region (shaded) in hi, showing the location of SEMA5A and a taste receptor gene (TAS2R1). e) UCSC Genome Browser view of the mouse VERT region (shaded) in mi, showing the location of Sema5a and a taste receptor gene (Tas2r119). f) UCSC Genome Browser view of the human VERT region (shaded) in hii above, highlighting the location of the AEI gene RETREG1. g) UCSC Genome Browser view of the mouse VERT region (shaded) in mii, showing the synteny between human and mouse for the protein coding genes in panel f, including Retreg1. h) VERT regions on human chromosome 16. The standard deviation (Std. Dev.) in 250 kb windows (circles) across human chromosome 16 is shown. Outlier windows are highlighted in different colors as shown. The red box and arrow mark the genomic location in panel i. i) UCSC Genome Browser view illustrating the genomic location represented in panel h, showing the location of GRIN2A (red box). j) VERT regions on mouse chromosome 16. The standard deviation in 50 kb windows (circles) is shown. Outlier windows from pre-B cell clones are highlighted in orange. k) UCSC Genome Browser view illustrating the genomic location of the VERT region (shaded area) represented in panel j above, with the location of Grin2a highlighted with a red box.

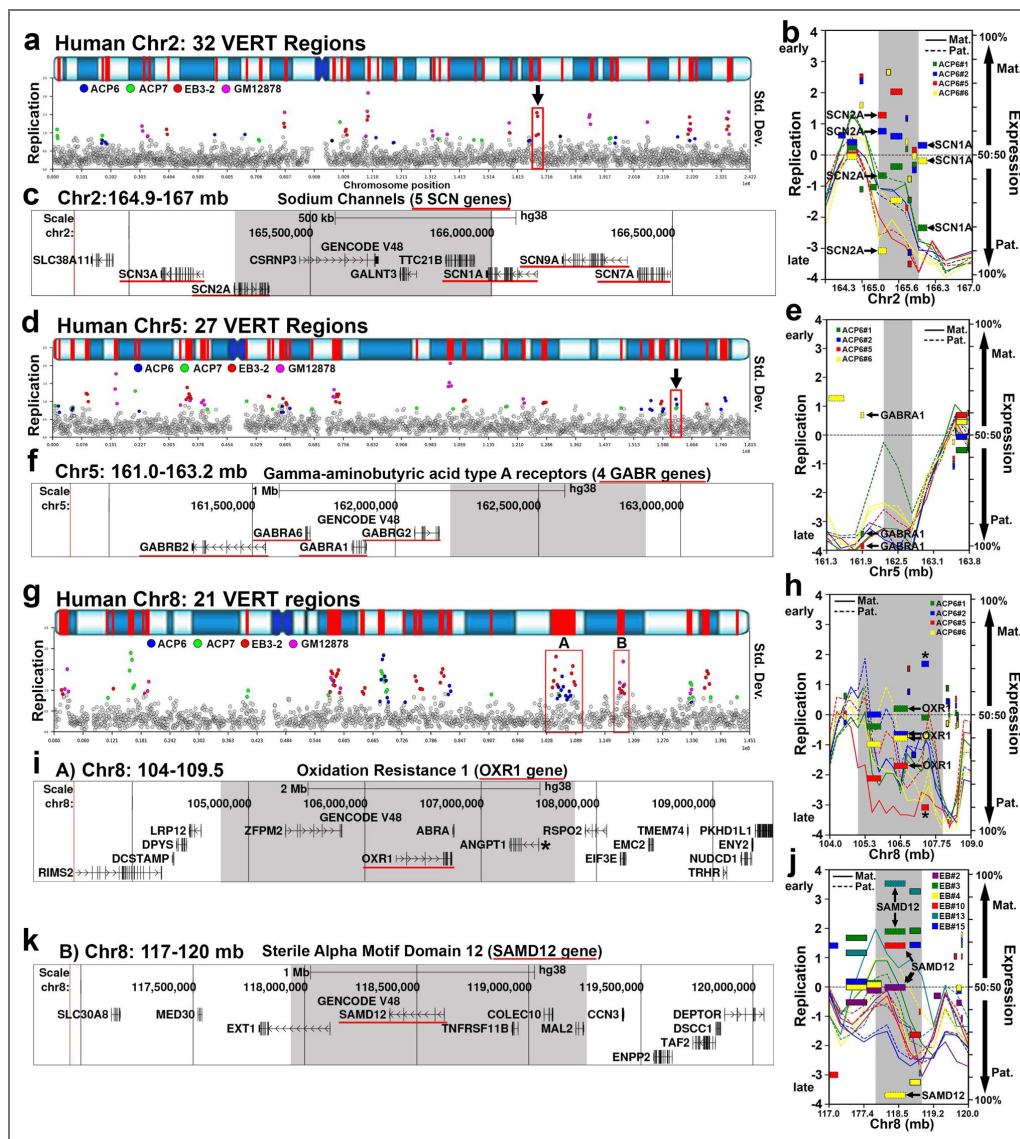


Figure 6. Epilepsy genes map to I/SCs.

a, d, and g) VERT regions on human chromosomes 2, 5, and 21. The top panels illustrate the G-banding (blue shading) pattern for each human chromosome, with VERT regions highlighted in red. The standard deviation (Std. Dev.) in 250 kb windows (circles) across each human chromosome is shown. Outlier windows from all four sets of clones (ACP6, ACP7, EB3-2, and GM12878) are highlighted in different colors as shown. The arrows and red boxes mark the VERT regions shown in panels b, c, e, f, h, i, j and k. b, e, h, j) Illustrations of I/SCs on chromosomes 2, 5, and 8. The shaded areas mark the VERT regions. Each clone was color coded to illustrate the VERT and AEI in different clones. The maternal (Mat.) and paternal (Pat.) alleles are indicated for replication and expression. c) UCSC Genome Browser view illustrating the genomic location of the VERT region (shaded area) represented in panels a and b, with the location of 5 sodium channel genes (SCN3A, SCN2A, SCN1A, SCN9A, and SCN7A) highlighted with a red line. e) Illustration of an I/SC on chromosome 5. The shaded area marks the VERT region. AEI of GABRA1 is shown. f) UCSC Genome Browser view illustrating the genomic location of the VERT region (shaded area) represented in panels d and e above, with the location of 4 GABA receptor genes (GABRB2, GABRB6, GABRA1, and GABRG2) highlighted with a red line. h) Illustration of a I/SCs (A) on chromosome 8. Each ACP6 clone was color coded to illustrate the VERT and AEI in different clones. The maternal (Mat.) and paternal (Pat.) alleles are indicated for replication and expression. The shaded area marks the VERT region. The location and expression of OXR1 is indicated. i) UCSC Genome Browser view illustrating the genomic location of the VERT region (shaded area) represented in panels g (A) and h above, with the location of the OXR1 gene highlighted with a red line. We note that ANGPT1 (marked with *), located within this I/SC displays AEI (see Supplementary Table 2). j) Illustration of an I/SC (B) on chromosome 8. AEI of SAMD12 is indicated. k) UCSC Genome Browser view illustrating the genomic location of the VERT region (shaded area) represented in panels g (B) and j above, with the location of the SAMD12 highlighted with a red line.

A second example of synteny is shown in Figure 5f–5g, where human RETREG1 and mouse Retreg1 are located within VERT regions. Human RETREG1 is a known AEI gene⁴⁵, and mutations in RETREG1 cause sensory and autonomic neuropathy²⁹. A third example is shown in Figure 5h–5k, where both human GRIN2A and mouse Grin2a are located within VERT regions, and loss of function mutations in GRIN2A are known to cause autosomal dominant (i.e. haploinsufficiency) epilepsy^{29,50}. These findings indicate that I/SC-associated epigenetic regulation is conserved across species, affects functionally related gene families, and may influence human genetic disease inheritance patterns (see below).

Neurodevelopmental disease genes map to I/SCs

To further investigate the relationship between I/SC regulation and human disease genes, we annotated the human protein-coding genes located within VERT regions (see Supplemental Table 4). This analysis identified 979 genes overlapping VERT regions, of which 272 are associated with single-gene disorders according to the OMIM database²⁹. Among these genes, 155 are linked to autosomal recessive diseases, 87 to autosomal dominant diseases, and 30 exhibit both dominant and recessive inheritance patterns (Supplemental Table 8). Consistent with our GO analysis, which revealed an enrichment for neurodevelopmental processes (see Fig. 1k), we found that various neurodevelopmental disease genes were prominently represented among the disease-associated genes (Table 3). Moreover, it was not unexpected to observe gene clusters associated with neurodevelopmental disorders residing within VERT regions. For example, Figures 6a–6c depict the location, I/SC profile, and genomic organization of a cluster of sodium channel genes (SCN family, $n = 5$). Mutations in SCN1A, SCN2A, SCN3A, and SCN9A genes are known to cause autosomal dominant developmental and epileptic encephalopathies, with haploinsufficiency as a primary mechanism (Table 3 and ^{29,51,52}). We also note that the syntenic locus in the mouse displays VERT, with the 5 syntenic mouse Scn genes residing within a VERT region on mouse chromosome 2 (Supplementary Fig. 2e–2f).

A second example of a cluster of epilepsy disease genes is shown in Figures 6d–6e, where four members of the gamma-aminobutyric acid type A receptor gene family (GABR family, $n = 4$) reside adjacent to a VERT region; notably, GABRA1 displays AEI (Fig. 6e, and Supplementary Table 2), and mutations in GABRA1, GABRB2, and GABRG2 cause autosomal dominant developmental and epileptic encephalopathies^{29,53,54}. We also note that a second GABR gene cluster ($n = 4$), where GABRA2 and GABRB1 are also associated with autosomal dominant developmental and epileptic encephalopathies, is adjacent to a VERT region on chromosome 4 (see Table 1). Two additional examples are shown in Figures 6g–6j, where OXR1 and SAMD12 are located in different VERT regions on chromosome 8, and mutations in each result in autosomal dominant epilepsy²⁹.

Another prominent group of human disease genes located within I/SCs are those associated with Parkinson disease (Table 3). As shown in Figure 7, four well-characterized Parkinson disease genes, SNCA, LRRK2, DNAJC6, and PARKN, are situated within or adjacent to VERT regions. Importantly, both SNCA and LRRK2 were among the first genes identified as displaying AEI in the study by Gimelbrant et al. in 2007⁴⁵. In our own dataset, we found that DNAJC6 exhibits AEI in LCL clones¹. We also highlight that JAK1, a gene involved in immune signaling, lies within the same I/SC as DNAJC6 (Fig. 7f). JAK1 is a known AEI gene^{55,56}, and mutations in JAK1 cause autosomal dominant autoinflammation, immune dysregulation, and eosinophilia (AIIDE). A recent study indicated that AEI of wild-type and mutant JAK1 alleles correlates with variable disease penetrance²⁸.

Taken together, these findings underscore the role of I/SCs in regulating disease-relevant gene clusters and emphasize their conservation, synteny, and functional relevance in both neurological and immune-related disorders. These examples highlight how stochastic epigenetic regulation within I/SCs contributes to phenotypic variability and disease risk.

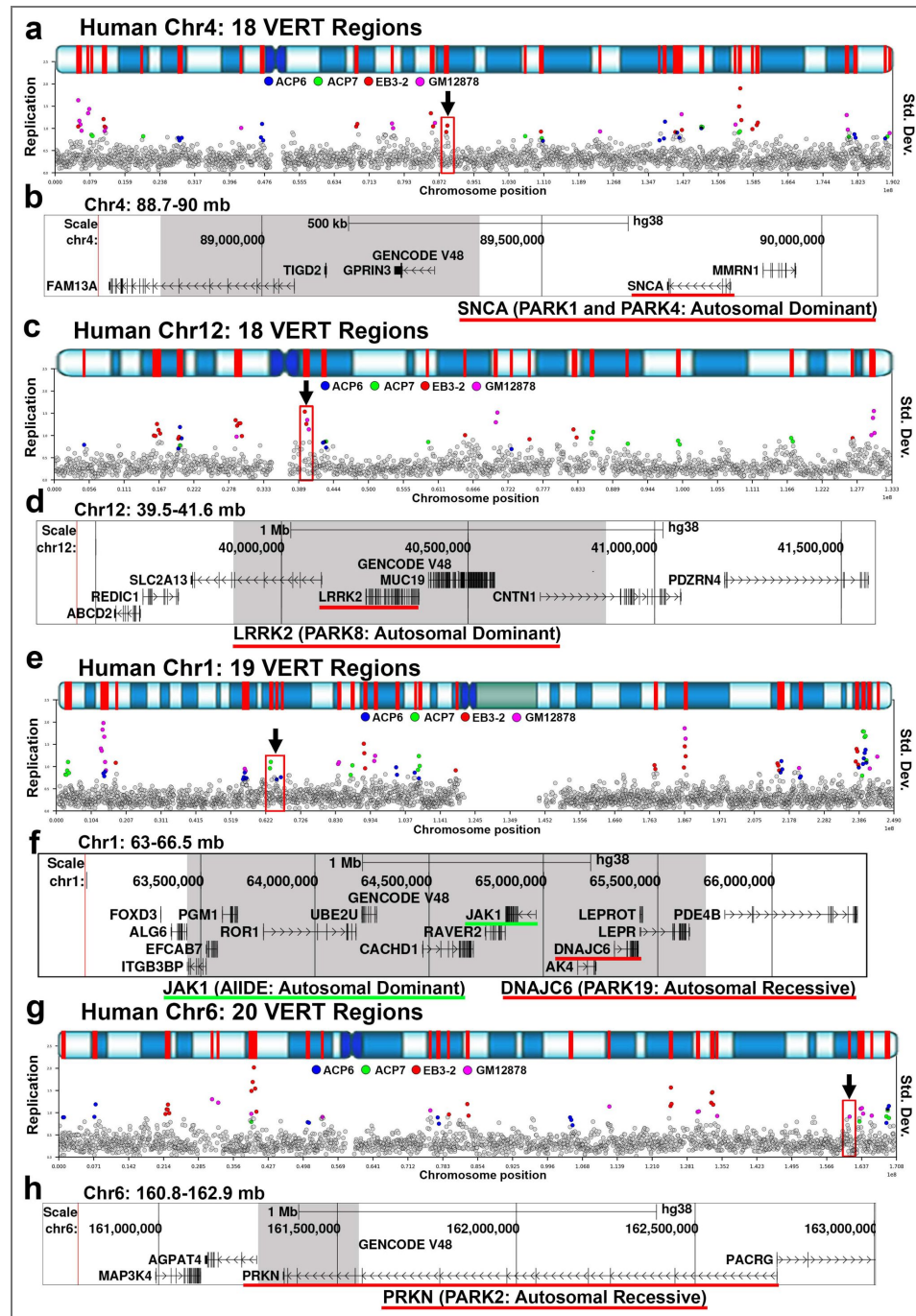


Figure 7. Parkinson disease genes map to I/SCs.

a, c, e, and g) VERT regions on human chromosomes 4, 12, 19, and 20, respectively. The top panels illustrate the G-banding (blue shading) pattern for each human chromosome, with VERT regions highlighted in red. The standard deviation (Std. Dev.) in 250 kb windows (circles) across each human chromosome is shown. Outlier windows from all four sets of clones (ACP6, ACP7, EB3-2, and GM12878) are highlighted in different colors as shown. The arrows and red boxes mark the VERT regions shown in panels b, d, f, and h, respectively. b) UCSC Genome Browser view illustrating the genomic location of the VERT region (shaded area) represented in panel a, with the location of the SNCA gene highlighted with a red line. d) UCSC Genome Browser view illustrating the genomic location of the VERT region (shaded area) highlighted in panel c, with the location of the LRRK2 gene highlighted with a red line. f) UCSC Genome Browser view illustrating the genomic location of the VERT region (shaded area) represented in panel e, with the location of the DNAJC6 gene highlighted with a red line; also highlighted with a green line is the JAK1 gene. h) UCSC Genome Browser view illustrating the genomic location of the VERT region (shaded area) represented in panel g, with the location of the PRKN gene highlighted with a red line.

Discussion

The co-occurrence of random monoallelic expression and asynchronous replication between alleles of autosomal genes is well documented^{2–4}, yet its developmental potential and contribution to human disease have not been systematically investigated. In this study, we expand on the discovery and characterization of Inactivation/Stability Centers (I/SCs), autosomal loci marked by AEI and VERT, by identifying an additional 163 VERT regions using clonal lines of primary ACP cells. This brings the total number of I/SCs to 363, covering approximately 12% of the human genome.

To assess the evolutionary conservation of I/SC regulation, we analyzed replication timing data from mouse pre-B cell clones, identifying 117 mouse VERT loci. Many of these regions overlap with known mouse AEI genes and gene clusters including olfactory, vomeronasal, and antigen receptors as well as protocadherin gene families. Notably, we identified 21 syntenic VERT regions shared between human and mouse genomes. These findings reinforce and extend previous observations indicating a high degree of conservation of AEI genes⁴ and suggest that I/SCs are a fundamental and widespread feature of autosomal epigenetic regulation in mammals.

I/SC regulation imparts cell-type specific and tissue-relevant variation

Our clonal analysis of ACP and LCL cells, combined with haplotype-phased replication timing and expression profiling, enabled us to identify tissue-specific I/SCs, while also confirming a subset of I/SCs that are shared across tissues and conserved between species. These findings support a dual regulatory model in which certain I/SCs may serve ubiquitous functions and are utilized broadly across cell types, whereas others are selectively engaged based on the gene expression program and functional requirements of a given tissue. This model is supported by hierarchical clustering analysis of VERT regions, which showed greater overlap within tissue type (ACP6 vs. ACP7; EB3-2 vs. GM12878) than between tissues, suggesting that I/SC activity is tightly linked to cellular identity.

Importantly, we identified 68 VERT regions shared across two or more clonal sets, representing high-confidence I/SCs that likely reflect core regulatory regions active across individuals and cell types (Fig. 1i [↗](#)). As detailed in Supplemental Table 1, we also identified VERT regions detected exclusively in single clone sets. These private VERT regions may reflect inter-individual variability in epigenetic regulation, context-dependent I/SC usage, but more likely reflect a limitation of sampling—i.e., regions that would be detected in additional clones if a larger number were analyzed. Thus, while our findings provide strong support for both shared and tissue specific I/SCs, they also underscore the importance of clone number and diversity in defining the full landscape of autosomal epigenetic regulation.

In the context of development, AEI introduces stochastic variability in gene dosage across clonal populations, contributing to cellular heterogeneity during lineage specification and tissue formation. Such heterogeneity could serve as a substrate for cellular selection, enabling the developing tissue to fine-tune function or respond adaptively to environmental cues. This model aligns with observations that AEI often affects genes involved in cell adhesion, signal transduction, and neurodevelopment, where even small variations in expression levels could significantly impact cell fate and connectivity.

At the molecular level, I/SCs contribute to cellular individuality by enabling stochastic monoallelic expression of specific genes. This mechanism is particularly critical in biological systems that require diversity at the single-cell level, such as the olfactory, vomeronasal, immune, and nervous systems. Consistent with this, many I/SCs coincide with gene clusters, including olfactory, vomeronasal, and antigen receptor genes, as well as HLA and protocadherin genes, all of which are known to undergo stochastic monoallelic expression. We identified numerous additional gene clusters residing within I/SCs, suggesting that these other gene families may also participate in tissue mosaicism and/or cellular individuality. Our current model proposes that I/SC regulation is integral to tissue development, where stochastic, epigenetically driven allelic regulation

introduces phenotypic variability among cells. This variability may promote functional diversity and selective advantage during development. Under this model, each tissue would engage a distinct but partially overlapping set of I/SC-regulated genes, aligned with its unique transcriptional and functional requirements. This framework positions I/SCs as key players in promoting cellular individuality, tissue mosaicism, and developmental plasticity.

Many of the genes that reside within I/SCs are regulated by allelic exclusion. Allelic exclusion is a regulatory mechanism by which only one allele of a gene is expressed while the other is actively silenced, ensuring monoallelic expression in each cell. This process is well characterized in the immune system, where it plays a critical role in B-cell and T-cell receptor gene rearrangements, ensuring that each lymphocyte produces a single antigen receptor ¹⁸. Furthermore, functional monoallelic expression is achieved via a secondary mechanism that involves product inhibition. For example, in developing B cells, expression of a productive immunoglobulin heavy chain allele suppresses rearrangement and expression of the second allele, maintaining clonal specificity ¹⁸. Allelic exclusion has also been observed in olfactory receptors, where product feedback ensures expression of a single allele of only one member of a large gene family, ensuring that each olfactory receptor neuron responds to a single odorant ^{2,33,57–60}.

I/SC regulation acts through mechanisms distinct from X inactivation

Numerous studies in both human and mouse cells support the notion that allele-specific epigenetic regulation of autosomal protein-coding genes is widespread ^{4,11,13,37,38,40,57,61,62}. Although stochastic monoallelic expression on autosomes may appear similar to X chromosome inactivation (XCI), the mechanisms behind them are distinct: First, XCI is initiated in preimplantation embryos, as early as the 4-8 cell stage in the mouse and 8-16 cell stage in humans ^{63–67}, and later becomes established during differentiation of the inner cell mass of the blastocyst in both species ^{67,68}. In contrast, autosomal stochastic monoallelic expression appears later in development, with allelic biases that can be erased and reestablished during differentiation ^{10,38}. For example, a recent study found that allele-specific autosomal inactivation is absent during early blood lineage specification but detectable in hematopoietic stem cells and their differentiated progeny ⁹. Second, in XCI, there is strong coordination along the chromosomes: genes on the active X chromosome share both expression and early replication profiles, whereas genes on the inactive X show silencing and late replication. In contrast, autosomal inactivation lacks such coordination: allelic expression and replication timing are unlinked within a chromosome and independent across the genome ^{1,7,69}. Taken together, these observations indicate that autosomal allelic inactivation operates via a mechanism distinct from XCI ^{1,9}.

An alternative model for haploinsufficiency

The relationship between genotype and phenotype is fundamental to understanding human genetic disease. At the scale of the whole genome, loss of function alleles are generally not deleterious when heterozygous and are therefore haplosufficient ^{21,24}. Not surprisingly, most of the loss of function mutations in single genes in humans are recessive ^{21,29}. Therefore, for most human disease genes a single normal allele is sufficient to achieve the typical phenotype. Dominantly inherited disorders comprise a diverse group of conditions that are often severe and poorly understood, except in cases of gain-of-function mutations, where dominant negative or interfering activities induce the abnormal phenotype. In addition, many dominantly inherited diseases result from copy number changes, either gain or loss of wild-type alleles ^{29,70}. Haploinsufficiency, predicted to apply to as much as 40% of the human genome ^{19,20}, refers to a heterozygous deletion or complete loss-of-function mutation that results in an altered phenotype, indicating that a single normal allele is insufficient. The underlying molecular mechanisms responsible for most haploinsufficient human disease genes remain largely unknown.

The existence of two distinct populations of cells in XX individuals, one expressing genes from the maternally derived X chromosome and the other from the paternally derived X, has profound implications for both development and the manifestation of X-linked diseases, where both

dominant and recessive inheritance patterns are observed in females²⁹. A key factor influencing these inheritance patterns is the presence or absence of cellular selection during development. In certain X-linked disorders, strong selection favors the survival or expansion of cells in which the mutant allele is on the inactive X chromosome, resulting in a predominance of wild-type cells and a clinical presentation consistent with X-linked recessive inheritance^{21,26}. In contrast, when cellular selection is absent, heterozygous females typically retain a ~50:50 mosaic of wild-type and mutant-expressing cells. The presence of mutant-expressing cells, essentially null cells, is thought to underlie X-linked dominant phenotypes^{27,71}. In such cases, heterozygous loss of function mutations leads to a condition of haploinsufficiency, in which the presence of a single normal allele is insufficient to maintain normal function. Interestingly, although single X-linked alleles are sufficient in males, the mosaic expression pattern in females unmasks phenotypic consequences that are otherwise hidden. This highlights an underappreciated mechanism of haploinsufficiency: the cellular mosaicism inherent to X inactivation can reduce effective gene dosage below functional thresholds in individuals with heterozygous mutations, particularly in tissues where cells expressing the mutant (null) allele persist.

One of the most striking implications of I/SC regulation is its relevance to human genetic disease, particularly those involving haploinsufficiency. Among the 979 genes mapped within human VERT regions, 272 are associated with monogenic diseases, including 87 with autosomal dominant inheritance, many of which are classified as haploinsufficient²⁹. Our findings offer a new mechanistic model for autosomal haploinsufficiency: in the absence of compensatory upregulation and/or cellular selection mechanisms, stochastic allelic inactivation results in or expands a functionally null cell population, even when one allele is wild type. This mirrors the situation observed in X-linked dominant disorders with loss of function mutations, where the absence of cellular selection results in the persistence of null cells. This model expands upon prior theories of haploinsufficiency that assume strict biallelic expression, which attribute dominant phenotypes to factors such as threshold-dependent expression, stoichiometric imbalance in multiprotein complexes, or inability to buffer expression noise^{22–25}. In addition, our model fits nicely with a previous study that modeled stochastic gene expression of disease genes and proposed “haploinsufficiency syndromes might result from an increased susceptibility to stochastic delays of gene initiation or interruptions of gene expression”⁷². Our data suggest that epigenetically regulated stochastic AEI can contribute to pathogenic phenotypes in heterozygous individuals by either reducing the number of functional cells below a critical disease threshold or by disrupting the normal mosaicism of gene expression within disease-relevant tissues. This expands the conceptual framework of haploinsufficiency to include epigenetic, rather than purely genetic, sources of dosage reduction.

Limitations of our study

While our findings provide substantial evidence for the existence and functional relevance of I/SCs, several important limitations remain to be addressed. First, the number of clones analyzed in our studies, while sufficient to identify robust patterns of AEI and VERT, limits our ability to fully capture the full spectrum of stochastic epigenetic states across the genome. For example, we frequently observe AEI of genes located adjacent to VERT regions, indicating that the boundaries of the epigenetic mechanisms functioning at I/SCs remain undefined. This raises important questions about how far the influence of I/SC-associated regulation extends beyond the VERT region. In addition, the specific epigenetic marks, including DNA methylation, histone modifications, and chromatin accessibility, at key regulatory elements such as replication origins, enhancers, and promoters within I/SCs have yet to be mapped. Furthermore, the higher-order chromatin organization of I/SCs, including their relationship to topologically associating domains (TADs) and A/B compartments^{49,73}, remains largely unexplored and may hold critical insights into their regulatory architecture and functional constraints. Second, our analysis was conducted in a restricted number of cell types, namely LCLs and ACPs, and therefore does not represent the full diversity of I/SC usage across human tissues. Third, our study was limited to a small number of individuals, which may constrain the identification of inter-individual variability in I/SC regulation. Fourth, the developmental timing of I/SC establishment remains unresolved; it is

unclear when during lineage commitment or differentiation these regions acquire their allele-specific states. A related and equally important question is whether I/SCs can be reset during reprogramming into induced pluripotent stem cells (iPSCs) and subsequently reestablished upon differentiation. This issue is particularly relevant given the widespread use of iPSCs in human disease modeling and regenerative medicine ^{74–76}. Understanding how I/SCs behave during cellular reprogramming and lineage specification will be critical for interpreting AEI-related phenotypes in iPSC-derived systems.

An additional open question is how the magnitude of stochastic AEI influences disease states, particularly in the context of variable penetrance and expressivity. For example, a recent study employed allelic expression ratios to identify functionally relevant AEI in genes associated with inborn errors of immunity, showing that individuals with the same pathogenic mutation can exhibit divergent clinical outcomes depending on the degree of allelic imbalance ²⁸. These findings suggest that AEI can modulate disease severity, acting as an epigenetic layer of regulation that determines whether a given mutation manifests clinically. We anticipate that the biological significance of AEI will vary depending on the function of the gene, its dosage sensitivity, and the cellular context in which it is expressed. Addressing these questions will be essential for achieving a comprehensive understanding of the formation, function, and disease relevance of I/SCs in human biology.

Conclusion

In conclusion, our work extends the catalog of I/SCs in the human genome and supports a model in which stochastic, allele-specific epigenetic regulation at autosomal loci contributes to cellular diversity and disease susceptibility. Moreover, by identifying syntenic I/SCs in the mouse genome, our findings greatly enhance the potential utility of mouse models for mechanistic studies of disease-causing mutations at conserved loci. Our findings present a potential mechanism for incomplete penetrance, variable expressivity, and tissue-specific vulnerability, particularly in disorders caused by haploinsufficiency. In such cases, stochastic silencing of the wild-type allele in a subset of cells may lead to functional null states, even in heterozygous individuals, reducing the effective number of functional cells below a critical threshold. This may be especially relevant in the nervous system, where precise gene dosage is often essential for maintaining neuronal function. Moreover, I/SC regulation at disease genes raises the possibility that stochastic epigenetic regulation could act as an epigenetic modifier of disease risk, helping to explain why some individuals carrying pathogenic mutations remain asymptomatic. Understanding the full scope of I/SC regulation in both normal development and disease pathogenesis will be essential for identifying novel regulatory mechanisms and potential therapeutic targets. As tools for single-cell and allele-resolved epigenomics continue to advance, I/SCs may emerge as key elements in both basic biology and precision medicine.

Methods

Resource availability

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Mathew Thayer (thayerm@ohsu.edu [✉](#))

Materials availability statement

All data generated or analyzed during this study are included in the manuscript and supporting files.

Cell culture

All human tissue samples were obtained in accordance with protocols approved by the Oregon Health & Science University Institutional Review Board (OHSU IRB# 00019018) and the Western Institutional Review Board (WIRB# 20193303). Informed consent was obtained from all donors or their legal guardians. ACP cells were grown in Dulbecco's Modified Eagle's Medium low glucose

(Life Technologies) supplemented with 10% fetal bovine serum (Hyclone). Single cell clones were isolated by plating individual cells in 6 well dishes, and were expanded for >20 population doublings⁷⁷. All cells were grown in a humidified incubator at 37°C in a 5% carbon dioxide atmosphere.

Haplotype Phasing of ACP cells

Short-read whole genome sequencing was obtained from two parent/child trios with a range of ~7-14x genome coverage. BWA⁷⁸ was used for alignment, and FreeBayes⁷⁹ used to call putative germline variants jointly for each trio. Variants were filtered to remove sites with Genotype Quality score of less than 30, and sites with missing genotypes for any member of the trio were removed. WhatsHap⁸⁰ was then used to phase variants in the child utilizing mendelian inheritance patterns and reads containing multiple variants. 1,679,207 high-confidence sites were phased in ACP7, and 837,323 sites were phased in ACP6.

Allele-specific Repli-Seq

Short read sequence data was aligned with BWA with standard parameters. BAM files were then deduplicated and sorted with SAMtools⁸¹. BCFtools mpileup⁸¹ was then used to produce allele-specific read counts at phased heterozygous loci. Library size normalization was performed to produce counts per million informative reads. Read counts were then grouped into 250kb sliding genomic windows with no overlap. E/L Repli-Seq values were generated by taking the Log base 2 of early versus late values⁸². Quantile normalization was performed on each group of subclones within each individual sample to allow for comparison among subclones. For each sample, variability of allele-specific RT among subclones was defined as the Standard Deviation (degrees of freedom= 1) of repli-seq at each 250kb window. Genome-wide std. dev. values generated a log-normal distribution, and we set a threshold of $Z > 2.25$ on the log-normal to identify regions with highly variable allele-specific RT among subclones.

Allele-specific RNA-seq

Nuclei were isolated by centrifugation for 0.5 minutes from ACP clones following lysis in 0.5% NP40, 140 mM NaCl, 10 mM Tris-HCl (pH 7.4), and 1.5 mM MgCl₂. Nuclear RNA was isolated using Trizol reagent using the manufacturer's instructions, followed by DNase treatment to remove possible genomic DNA contamination. Briefly, ribosomal RNAs were removed using the Ribo-Zero kit (Illumina), RNA was fragmented into 250-300bp fragments, and cDNA libraries were prepared using the Directional RNA Library Prep Kit (NEB). Paired end sequencing was done on a NovaSeq 6000 at the OHSU MPSSR core facility. Strand specific short read sequence data was aligned with STAR⁸³. SAMtools⁸¹ was used to remove duplicates and sort BAM files. Strand-specific, allele-specific reads at heterozygous phased SNPs were counted using BCFtools mpileup⁸¹ and grouped together for all SNPs within each gene. Allelic expression imbalance was calculated as the fraction of reads derived from the maternal vs the paternal allele. For example, completely balanced allelic expression is 50% AEI, and completely imbalanced AEI is 100%. A threshold of AEI $\geq 70\%$, and FDR corrected binomial p-value ≤ 0.05 was used to classify AEI in each individual subclone. Standard deviation (degrees of freedom=1) was used for each gene to generate a metric of AEI variability among subclones. A genome-wide distribution of std. dev. was generated across all genes, and the top 2.5% of variable genes were classified as highly variable among subclones.

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Additional information

Author Contribution Statement

MBH, MJT, and DMG conceived of the experimental design for the RNA expression and replication timing assays. KF performed the surgeries to obtain the amputated digits. BJ and PAY performed the ACP cultures and clone generation. MBH and AEV carried out the RNA-seq assays, including data analysis. MBH and AEV carried out the Repli-seq assays, including data analysis. PFC and PTS provided advice during preparation of the manuscript. MBH, DMG and MJT wrote the manuscript.

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Additional files

Table 1. [VERT regions at gene clusters in the human and mouse genomes.](#)

Table 2. [VERT regions showing synteny between human and mouse genomes.](#)

Table 3. [Neurodevelopmental genes within VERT regions.](#)

Supplemental Figures. [VERT regions at gene clusters in the human and mouse genomes.](#)

Supplemental Table 1. [VERT regions showing synteny between human and mouse genomes.](#)

Supplemental Table 2. [Neurodevelopmental genes within VERT regions.](#)

Supplemental Table 3. [VERT regions at gene clusters in the human and mouse genomes.](#)

Supplemental Table 4. [VERT regions showing synteny between human and mouse genomes.](#)

Supplemental Table 5. [Neurodevelopmental genes within VERT regions.](#)

Supplemental Table 6. [VERT regions at gene clusters in the human and mouse genomes.](#)

Supplemental Table 7. [VERT regions showing synteny between human and mouse genomes.](#)

Supplemental Table 8. [Neurodevelopmental genes within VERT regions.](#)

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Peer reviews

Reviewer #1 (Public review):

Summary:

The existence of VERT regions is well supported, but the number of regions called as ISCs may be inflated by permissive thresholds (e.g., AEI {greater than or equal to} 0.8 or {less than or equal to} 0.2 in a single clone). This risks conflating transient stochastic differences with stable ISCs. Similarly, the claim of cell-type specificity is not convincingly demonstrated given the small sample size (n=4) and strong batch confounding between lymphoblastoid and cartilage progenitors. While syntenic VERT regions across mouse and human are intriguing, they complicate interpretation of strong clustering by cell type. Sampling depth may also have exaggerated allelic imbalance calls.

The proposed role of ISCs in haploinsufficiency is conceptually interesting but remains speculative; developmental stochasticity and founder population size may play larger roles than replication timing. The claim that autosomal inactivation is mechanistically distinct from XCI, however, is reasonable and supported.

Some conclusions should be more explicitly qualified as preliminary. Cell-type specificity and mitotic stability both require stronger evidence; the latter is inferred indirectly from clonal expansion rather than shown directly, and orthogonal experiments (e.g., allele-specific ChIP-seq, DNA methylation) would be required. Estimated genomic coverage of ISCs should also be re-evaluated, as single-clone observations may inflate counts.

Replication is limited. Hierarchical clustering is confounded by batch and based on presence/absence calls that lack quantitative resolution. More robust approaches would include using magnitude of imbalance, annotating VERTs by genomic location, applying stricter thresholds for replication timing, and benchmarking AEI distributions against the X chromosome. These are realistic re-analyses requiring no new data and could be completed in ~1 month.

Methods are generally well described and reproducible. Figures and text would benefit from improved clarity: axis labels are missing in places (e.g., Fig. 1c, Fig. 2g), legends should explain chromosome arm colors, and cluttered figures such as Fig. 1j could be re-visualized

for interpretability. Gene set enrichment analysis should be restricted to avoid inflated significance from overly broad categories. A useful citation for XCI timing (pmid=39420003) could be added to strengthen background.

Significance:

Conceptually, this work introduces ISC-like phenomena in human and mouse progenitor lines, coupling allelic expression imbalance with replication timing. Technically, it combines allele-specific RNA-seq with Repli-seq in genotyped, clonal, single-cell-derived lines. Clinically, it suggests an alternative model for haploinsufficiency, relevant to dosage-sensitive diseases where stochastic transcriptional delays could shape penetrance.

The study builds on prior work in allelic exclusion (e.g., HLA, olfactory receptors) and random monoallelic expression, generalizing these phenomena into ISC/vert frameworks and proposing mitotic stability of allele choice. By extending beyond expression to replication timing, the authors suggest a broader paradigm for epigenetic regulation at autosomal loci.

The paper will be of interest to epigeneticists studying XCI, allelic exclusion, and monoallelic expression; to developmental biologists examining replication timing and differentiation; and to clinicians concerned with dosage-sensitive and haploinsufficient disorders.

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Reviewer #2 (Public review):

Summary:

- This is a complicated research topic that touches on a few sub-fields of biology, and thus to make the paper more approachable I would recommend a careful edit of the text for clarity and precision of language.
- Authors point out that this is a decades-old field; it would make sense to use terminology established within the field rather than inventing their own. Allelic imbalance has been referred to as AI, MAE (monoallelic expression), RMAE (random monoallelic expression) etc. The paper whose mouse data the authors make use of uses Asynchronous Stochastic Replication Timing (ASRT) instead of VERT to refer to the same phenomenon. Creating unnecessary jargon makes the paper more difficult to read and adds needless complexity to an already complex field.
- Methods do not provide sufficient detail to fully evaluate or reproduce these experiments.
- It is helpful to show representative loci as the authors do in Fig 1F and G and Fig 2, but these panels are very densely rendered and thus difficult to process visually - even the cartoon version (1D) is thick with overlapping lines. The point that allelic imbalance is enriched in VERTs would be enhanced if the authors could present the allelic ratio for all genes found in all VERTs, demonstrating how replication timing on either chromosome affects the allelic ratio.
- The authors make the important point that VERTs are unlikely to be shared among different cell types and tissues (Fig 1i) but then find an enrichment for neuronal and immune genes in VERT regions identified in ACPs. It follows that these same genes are unlikely to be in such regions in the tissues where they are relevant. Some of the GO terms presented are too broad to suggest any biological significance to the result, even if there is statistical significance (for example, the top term for LCL clones 'Cytoplasm' is associated with 12,000 genes, and the second term for mouse clones 'Membrane' is associated with 10,000). It would be helpful to focus on GO terms lower in the GO hierarchy.

- Figure 3 highlights the association of related gene clusters with VERTs but the VERTs are assigned based on variable replication timing in just 1 or 2 clones. This is an interesting observation, but to make the point that "VERT regions frequently coincide with gene clusters in the human genome" there needs to be a systematic assessment of replication timing at all gene clusters across all clones, and a statistical test for significance.
- It is an interesting hypothesis that VERTs are conserved between species at syntenic loci. If such regions are really conserved, one would expect that replication timing at these sites would be consistently asynchronous. However, the data presented shows that in human clones these VERTs can be specific to an individual donor (as in 5A) or an individual clone (as in 5H).
- Again, the finding that VERTs coincide with neurodevelopmental disease genes in immune and cartilage cells is at odds with the previous statements and data about the tissue specificity of VERTs. In order to support the claim that neurodevelopmental disease associated genes reside in asynchronously replicating regions, and are thus more prone to allelic imbalance, the authors would need to demonstrate this phenomenon in neuronal cells.

Significance:

The authors pair analysis of replication timing and allele-specific expression in clonal populations of primary human cells. They combine these data with previously published data on clones from transformed human cell lines. They identify a number of genomic regions that display asynchronous replication timing in at least one clone and correlate these regions with allele-specific expression of genes within them. They also observe that several interesting gene sets, including genes that are associated with human diseases, map to asynchronously replicating regions. This is a good experimental approach that builds on already published data demonstrating the connection between allelic imbalance and replication timing. However, the authors consistently lean on thin evidence (i.e. a single clone) within a modestly sized dataset (4 clones from 2 donors each) to propose a new model for haploinsufficiency in human disease. The consistent focus on limited elements in the data and perhaps an overreach in the interpretation makes it difficult to appreciate what is in fact a very good experiment.

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Author response:

General Statements

We thank the reviewers for their thoughtful and constructive comments, which will substantially improve our manuscript. In response, we will revise the text and figures throughout to address the points raised. Specifically, we will:

- i. Refine our definition of Inactivation/Stability Centers (I/SCs): We will limit this designation to loci where both Allelic Expression Imbalance (AEI) and Variable Epigenetic Replication Timing (VERT) are detected, either in the present study or in previously published work.
- ii. Expand methodological clarity: We will provide detailed descriptions of how VERT regions were identified, annotated, and quantified, including thresholds for allelic imbalance, replication timing variability, and sampling depth. We also justify the $\geq 80\%$ AEI cutoff, which is based on recent studies showing that modest allelic biases can have biological and clinical significance.
- iii. Enhanced benchmarking and validation: In addition to the analysis of X inactivation in female ACP cells, we will include comparisons between imprinted and non-imprinted regions to benchmark the magnitude of allelic replication timing imbalance, demonstrating that the

magnitude of imbalance observed at imprinted loci is comparable to that at the non-imprinted VERT regions.

iv. Address tissue specificity and sampling limitations: We will discuss the limited number of clones, tissues, and individuals analyzed, emphasizing that while our data identify robust AEI and VERT patterns, additional tissues and individuals will be required to capture the full diversity of I/SC regulation.

v. Clarify biological relevance: We will expand our discussion to highlight the consistency of AEI findings across cell types, including examples of genes implicated in neurodevelopmental and neurodegenerative disorders, and we will clarify our model of how I/SC regulation may contribute to haploinsufficiency, variable expressivity, and incomplete penetrance in human disease.

vi. Improved figures and supplemental data: We will update figure legends for clarity, add a new supplementary figure comparing imprinted and non-imprinted regions, and cross-reference all supplemental tables.

We believe these revisions strengthen the manuscript conceptually and experimentally, and we thank the reviewers and editors for their valuable feedback.

Description of the planned revisions

Reviewer #1:

The existence of VERT regions is well supported, but the number of regions called as ISCs may be inflated by permissive thresholds (e.g., AEI {greater than or equal to} 0.8 or {less than or equal to} 0.2 in a single clone). This risks conflating transient stochastic differences with stable ISCs.

We selected the >80% (or <20%) allelic imbalance threshold, along with the requirement of at least one biallelic clone, as our criterion for significant AEI. This choice was guided by a recent study demonstrating that allelic imbalance as low as a 65%/35% is enough to effect disease penetrance in humans (Nature 2025; 637:1186–1197). For completeness, results obtained using more stringent thresholds (>90% and >95% imbalance) are presented in Supplementary Table 2.

Furthermore, it is unlikely that transient stochastic differences in allelic expression, such as those detected by single-cell RNA sequencing assays (Nat. Rev. Genet. 2015; 16:653–664), would be captured by our approach. Each clone in our study was expanded from a single cell to over one million cells before both RNA-seq and Repli-seq analysis, effectively averaging out transient transcriptional and/or replication fluctuations, and thus reflecting stable, mitotically heritable epigenetic states.

More robust approaches would include using magnitude of imbalance, annotating VERTs by genomic location, applying stricter thresholds for replication timing, and benchmarking AEI distributions against the X chromosome.

All VERT regions identified in this study were annotated according to both the magnitude of allelic imbalance and their genomic coordinates, using 250 kb windows for the human samples and 50 kb windows for the mouse samples (see Supplementary Tables 1 and 6). Figure 1c directly compares the magnitude of imbalance, defined as outliers in the standard deviation, for both allelic replication timing and allelic expression across autosomal and X-linked loci in female ACP cells.

In addition, we will benchmark the magnitude of replication timing imbalance using autosomal imprinted regions as a second internal control. We detected allelic replication imbalance at 13 known imprinted loci, and the standard deviation of replication timing at

these loci, measured in 250 kb windows, is comparable to that observed across the >350 VERT regions detected at non-imprinted sites. To illustrate this comparison, we will include a supplementary figure directly comparing imprinted and non-imprinted regions.

Figures and text would benefit from improved clarity: axis labels are missing in places (e.g., Fig. 1c, Fig. 2g), legends should explain chromosome arm colors, and cluttered figures such as Fig. 1j could be re-visualized for interpretability.

Figure labels will be added to Figs. 1c and 2g, and legends will be modified for clarity.

“the claim of cell-type specificity is not convincingly demonstrated given the small sample size (n=4) and strong batch confounding between lymphoblastoid and cartilage progenitors.” And “Hierarchical clustering is confounded by batch and based on presence/absence calls that lack quantitative resolution.”

We agree that the limited number of individuals and clones, as well as the comparison between only two distinct tissue types (LCLs and ACPs), have quantitative limitations. Our primary intent was to evaluate whether any I/SCs were shared between independently derived clonal datasets and to determine whether there is evidence of tissue-specific I/SC usage, rather than to make quantitative claims about global cell-type specificity.

To address this concern, we will replace the hierarchical clustering analysis currently shown in Figure 1i with a Venn diagram that more directly illustrates the overlap and tissue-specific distribution of VERT regions detected in the different clonal sets. This revised representation avoids assumptions about clustering relationships and removes batch-driven bias, while still conveying the key observation that many VERT regions are shared across tissues and others appear tissue-restricted.

While syntenic VERT regions across mouse and human are intriguing, they complicate interpretation of strong clustering by cell type. Sampling depth may also have exaggerated allelic imbalance calls.

We note that the human LCLs used in our study are B cells, and immunoglobulin gene rearrangements were used to confirm the clonal uniqueness of each line. Similarly, the mouse replication timing data analyzed here was generated from pre-B cells, which also undergo immunoglobulin gene rearrangement. Thus, both the human LCL and mouse pre-B cell datasets were derived from B-cell lineages, providing a consistent cellular context for comparative analysis.

Sequencing depth is an important consideration for all variant base calls. Without fully haplotype-resolved genomes, previous studies relied on calculating per-SNP calls of allelic imbalance based on reads covering a single nucleotide locus. To improve sequencing depth supporting the identification of VERT and AEI regions, we utilized fully haplotype-resolved genomes that allowed all informative allele-specific reads to be pooled across all heterozygous SNPs within genomic windows or expressed genes. For AEI, we set a minimum threshold of 20 informative allele-specific reads per gene, a minimum FDR-corrected p-value of ≤ 0.05 , and a minimum of 80% vs 20% allelic imbalance. Importantly, a recent study has shown that allelic imbalance as low as a 65%/35% is enough to effect disease penetrance in humans (Nature 2025; 637:1186–1197). We reiterate that more stringent thresholds (>90% and >95% imbalance) are presented in Supplementary Table 2.

Gene set enrichment analysis should be restricted to avoid inflated significance from overly broad categories.

Reviewer #2:

Some of the GO terms presented are too broad to suggest any biological significance to the result, even if there is statistical significance (for example, the top term for LCL clones 'Cytoplasm' is associated with 12,000 genes, and the second term for mouse clones 'Membrane' is associated with 10,000). It would be helpful to focus on GO terms lower in the GO hierarchy.

We will include our complete Gene Ontology analysis, with more specific biological categories, in Supplemental Table 5.

Allelic imbalance has been referred to as AI, MAE (monoallelic expression), RMAE (random monoallelic expression) etc. The paper whose mouse data the authors make use of uses Asynchronous Stochastic Replication Timing (ASRT) instead of VERT to refer to the same phenomenon. Creating unnecessary jargon makes the paper more difficult to read and adds needless complexity to an already complex field.

While we agree that allelic expression imbalance has been described by different investigators using many different phrases, we believe that MAE, RMAE and AI do not represent an accurate description of the phenomenon. In our study [and our previous study; Nat Commun. 2022; 13(1):6301] we used clonal analysis of allele-specific expression and found that while some clones display equivalent levels of expression between alleles of a given gene (i.e. bi-allelic expression) other clones express only one allele (i.e. mono-allelic expression), and yet other clones have undetectable expression (i.e. silent on both alleles). This pattern of allele-restricted expression indicates that each allele independently adopts either an expressed or silent state. Importantly, because these expression states are mitotically stable, allele-autonomous, and independent of parental origin, we refer to the choice of the expressed allele as stochastic. Given this variability, we believe that the phrase “Allelic Expression Imbalance” (AEI) represents a more accurate descriptor for this phenomenon. We also point out that “Allelic Expression Imbalance” has been used >120 times in the Pubmed database.

In addition, the replication asynchrony that exists at these loci is not consistent with purely ASynchronous Replication Timing (ASRT) between alleles. We found that each allele can independently adopt either earlier or later replication timing in different clones. This variability results in some clones exhibiting pronounced asynchrony between alleles, while in others, the two alleles replicate synchronously, with both adopting either the earlier or later timing state. As reported in our previous study (Nat. Commun. 2022; 13:6301), this behavior reflects a stochastic and allele-autonomous process, leading us to describe these loci as exhibiting Variable Epigenetic Replication Timing (VERT), which we believe is a more accurate descriptor of this phenomenon.

The point that allelic imbalance is enriched in VERTs would be enhanced if the authors could present the allelic ratio for all genes found in all VERTs, demonstrating how replication timing on either chromosome affects the allelic ratio.

The stochastic nature of allelic expression and replication timing observed at VERT loci indicates that each allele independently acquires its epigenetic state. Specifically, the expressed or silent status of one allele does not predict the replication timing or expression status of the opposite allele. Accordingly, the Early/Late pattern of replication timing that we detect, both in this study and in our previous work (Nat. Commun. 2022; 13:6301), is not correlated with which allele is transcriptionally active. This supports our conclusion that asynchronous replication timing is not a downstream consequence of monoallelic transcription, but rather an independent epigenetic feature of I/SCs. Regardless, we will provide the combined expression ratios for all transcripts that are located within the VERT regions in a Supplemental Table.

In addition, our analysis of imprinted loci reveals that even at genomic regions with parent-of-origin-specific expression, replication timing does not align with allelic activity: both early- and late-replicating alleles can be transcriptionally active, depending on the gene. This observation is consistent with the complex organization of many imprinted domains, where genes on opposite alleles exhibit reciprocal expression patterns. To illustrate this point, we will include a new supplemental figure demonstrating that imprinted loci harbor genes expressed from both the earlier- and later-replicating alleles.

Figure 3 highlights the association of related gene clusters with VERTs but the VERTs are assigned based on variable replication timing in just 1 or 2 clones. This is an interesting observation, but to make the point that "VERT regions frequently coincide with gene clusters in the human genome" there needs to be a systematic assessment of replication timing at all gene clusters across all clones, and a statistical test for significance.

Our intent in Figure 3 was not to suggest that all gene clusters are subject to VERT and AEI, but rather to highlight that several well-characterized multigene families that are known to exhibit random AEI, such as olfactory receptor and HLA gene clusters, coincide with VERT regions at their genomic locations. These examples serve as representative illustrations demonstrating that I/SC-associated regulation occurs at established AEI loci organized in gene clusters.

To clarify this point, we will revise the text to explicitly state that Figure 3 presents illustrative examples of known AEI-associated gene clusters overlapping with VERT regions, rather than a comprehensive or statistically exhaustive analysis of all gene clusters across the genome.

It is an interesting hypothesis that VERTs are conserved between species at syntenic loci. If such regions are really conserved, one would expect that replication timing at these sites would be consistently asynchronous. However the data presented shows that in human clones these VERTs can be specific to an individual donor (as in 5A) or an individual clone (as in 5H).

As discussed in our Limitations section, our analysis was restricted to a limited number of cell types, clones, and individuals, which may not capture the full diversity of I/SC usage across tissues and populations. While our dataset was sufficient to identify robust patterns of AEI and VERT, it likely represents only a subset of the broader landscape of I/SC regulation in both humans and mice. We anticipate that future studies incorporating a wider range of tissues, individuals, and clonal analyses will uncover an even greater degree of conservation and diversity in I/SC usage across genomes.

In order to support the claim that neurodevelopmental disease associated genes reside in asynchronously replicating regions, and are thus more prone to allelic imbalance, the authors would need to demonstrate this phenomenon in neuronal cells.

We make two points that address this critique: First, many of the neurodevelopmental disease genes located within or adjacent to VERT regions are not exclusively expressed in neuronal cells and have already been shown to exhibit AEI in non-neuronal contexts. For example, Gimelbrant and Chess (Science, 2007; 318:1136–1140) demonstrated AEI of the Parkinson disease genes SNCA and LRRK2 in lymphoblastoid cell lines (LCLs), and in our previous study, we detected AEI of DNAJC6, another Parkinson disease gene, in LCL cells (Nat. Commun. 2022; 13:6301). In the present study that used ACP cells, we identified VERT and AEI of several epilepsy-associated genes, including SCN1A, SCN2A (Fig. 6b), GABRA1 (Fig. 6e), and SAMD12 (Fig. 6j), as well as a gene implicated in autism and neurodevelopmental disorders, SEMA5A (Fig. 5c).

Second, independent studies from the E. Heard laboratory have provided further evidence that AEI occurs in neuronal lineages. Using mouse neural progenitor cells (NPCs), they identified genes subject to AEI (Dev. Cell, 2014; 28:366–380) and they later evaluated AEI of syntenic human neurodevelopmental disease genes, including Snca, App, Eya4, and Grik2 (Nat. Commun. 2021; 12:5330). In addition, they used the phrase “Allelic Expression Imbalance” to describe the epigenetic expression biases at these genes.

Together, these findings reinforce that AEI, and by extension I/SC regulation, is not restricted to specific cell types, but rather represents a generalizable mechanism of stochastic epigenetic regulation that includes genes relevant to neurodevelopment and disease.

However, the authors consistently lean on thin evidence (i.e. a single clone) within a modestly sized dataset (4 clones from 2 donors each) to propose a new model for haploinsufficiency in human disease. The consistent focus on limited elements in the data and perhaps an overreach in the interpretation makes it difficult to appreciate what is in fact a very good experiment.

We agree that our analysis was conducted on a modest number of clones and individuals, which we explicitly acknowledge as a limitation of the present study. However, several key points support the robustness and broader relevance of our conclusions:

- i. Clonal Design and Replication: The strength of our approach lies in its clonal resolution. Each clone represents a single-cell–derived population expanded to over a million cells, enabling direct detection of stable, mitotically heritable allele-specific epigenetic states that would not be apparent in population-averaged data. Importantly, many of the VERT regions we identified are shared between independent clones from different donors and across distinct cell types (ACP and LCL), demonstrating reproducibility and biological consistency.
- ii. Cross-Species Validation: We further identified syntenic VERT regions in mouse pre-B cell clones, including at loci known to exhibit AEI in prior studies, providing independent validation and evolutionary conservation of the phenomenon.
- iii. Integration with Published Evidence: Our findings extend prior observations of AEI and variable replication timing (e.g. Gimelbrant et al. Science 2007; Heskett et al. Nat. Commun. 2022) and are fully consistent with known stochastic allelic expression imbalance of autosomal genes. We also draw parallels with the absence of cellular selection mechanisms that dictate dominant inheritance patterns for loss of function alleles for X linked disease genes (reviewed in: J Clin Invest, 2008, 20-23; and Nat Rev Genet. 2025, 26, 571–580). Our proposed model linking I/SC regulation to haploinsufficiency is therefore a synthesis of our results with an extensive body of published data, not an inference drawn from isolated observations.
- iv. Scope and Framing: We will revise the manuscript to clarify that our proposed model represents a mechanistic framework, not a definitive or exclusive explanation, for how stochastic allelic regulation could contribute to dosage-sensitive disease phenotypes. We will also explicitly discuss the need for larger datasets and additional tissues to refine and test this model.

In summary, while we recognize the limited sampling inherent to clonal analyses, the consistency of our observations across donors, cell types, and species, together with prior corroborating studies, supports the validity of the conclusions and justifies the broader conceptual implications.

Description of analyses that authors prefer not to carry out

Reviewer #1:

Cell-type specificity and mitotic stability both require stronger evidence; the latter is inferred indirectly from clonal expansion rather than shown directly, and orthogonal experiments (e.g., allele-specific ChIP-seq, DNA methylation) would be required.

We disagree with this reviewer that the mitotic stability of the epigenetic states are “inferred indirectly from clonal expansion rather than shown directly”. Our experimental design inherently captures mitotically stable, allele-specific states because each clonal line is derived from a single progenitor cell and expanded to millions of cells before analysis. The allele-specific replication timing and expression profiles observed in these clones therefore reflect epigenetic states that are stably inherited across many cell divisions, rather than transient or stochastic fluctuations. This approach was also validated in our previous study (Nat. Commun. 2022; 13:6301), where the same clonal strategy demonstrated stable allele-restricted replication and expression patterns over extended passages.

We agree that orthogonal assays such as allele-specific ChIP-seq or DNA methylation analyses would provide additional mechanistic detail on the nature of I/SC-associated regulation. However, these experiments fall outside the scope of the present study, which was designed specifically to identify and map autosomal loci that exhibit coordinated AEI and VERT, the defining epigenetic features of I/SCs. While we fully acknowledge that defining the precise molecular marks (e.g., histone modifications, DNA methylation, chromatin accessibility) that underlie I/SC regulation will be an important future direction, our current data provide a genome-wide, allele-resolved foundation upon which such mechanistic studies can build.

In summary, the current dataset achieves the central goal of defining the genomic distribution and conservation of I/SCs based on functional readouts of replication timing and expression. Future work will extend these findings using allele-specific epigenomic profiling to characterize the epigenetic modifications associated with I/SC stability and cell-type specificity.

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